CS 411 Lab 2 (Individual)

Product Specification

Phillip T. De Guzman, Team Silver

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1 Introduction

This section outlines the purpose, scope, definitions, acronyms, abbreviations, references, and overview of BevOS.

1.1 Purpose

The purpose of this document is to define the scope, explain definitions, acronyms, abbreviations, cite references, and show an overview of BevOS.

1.2 Scope

Beverage manufacturers still coordinate production through spreadsheets, email, and disconnected systems. These manual processes create communication delays and approval bottlenecks as companies grow. With the beverage packaging market projected to reach $222.1 billion by 2030 (Grand View Research, 2024), that complexity breeds tension and miscommunication between parties, raising the need for coordinated, scalable information systems across multiple stakeholders (Zimmerman, 2026).

The needed solution consolidates operational activity into one platform with real-time visibility into requests, approvals, schedules, production progress, and delivery status. It tracks approvals automatically, schedules production milestones so delays surface before they affect delivery, links production completion to shipping and delivery, stores documents where authorized users find the current version, and turns operational data into dashboards for oversight. Role-based access limits each user to appropriate information and actions, and the system scales with business growth without being replaced.

BevOS is a centralized workflow management platform for beverage manufacturers, connecting brands, suppliers, distributors, co-packers, logistics providers, and internal operations in one system. A functional prototype will demonstrate how its interface, services, modules, and database work together to solve the coordination problem.

1.3 Definitions, Acronyms, and Abbreviations

The following are some definitions that are used throughout this document related to the project (Team Silver, 2026).

● Approval Workflow - A process used to review and approve operational tasks before they move to the next stage.

● BevOS - A centralized workflow-management platform designed for beverage manufacturing operations.

● Client - An organization or individual purchasing services or products from another participant in the manufacturing process.

● Co-Packer - A third-party company that manufactures and packages beverages for another company.

● Core Services - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.

● Dashboard - A visual interface displaying operational information and performance metrics.

● Distributor - An organization responsible for delivering products from manufacturers to retailers or customers.

● ERP (Enterprise Resource Planning) - Software that integrates core business processes into a unified system.

● Logistics - The planning and coordination of transportation and delivery activities.

● MFCD (Major Functional Components Diagram) - A diagram illustrating the primary hardware and software components of the system.

● Nice Package Company - The beverage-industry client and domain partner for whom BevOS is being developed.

● PostgreSQL - An open-source relational database management system used to store application data.

● Role-Based Access Control - A security method that grants system permissions according to the user's assigned role and responsibilities.

● Single Source of Truth - A trusted, centralized set of current information used consistently by authorized stakeholders.

● Stakeholder - A person or organization that participates in or is affected by a beverage-manufacturing workflow.

● Workflow - A sequence of business activities completed to achieve a specific objective.

1.4 References

● Grand View Research. (2024). Beverage Packaging Market Size, Share & Trends Analysis Report.

● Kennedy, T. (2026). CS410 Professional Workforce Development.

● Wilson, T. (2026). BevOS Presentation / Nice Package Co. Guest Presentation.

● Zimmerman, S. (2026) How the food and beverage industry is rethinking manufacturing investments. Food Dive.

● Team Silver. (2026, October 26). Lab 1 - BevOS Product Description. Retrieved October 26, 2026 from https://jeweliana.github.io/silver-bevOS/presentation.html

1.5 Overview

Section 2, Overall Description, will describes what the team is building and what components will be a part of the project. It will cover product perspective, product functions, user characteristics, constraints, assumptions, and dependencies.

Section 3, Specific Requirements, converts that description into verifiable statements. It will state functional requirements grouped by capability area with stable identifiers and priority marking. It will cover quantifying performance targets, records any additional assumptions and constraints surfaced during specification, and state any non-functional requirements for security, reliability, usability, maintainability, portability, and testability.

The appendix will include additional information such as supporting materials, and a list of equipment and software required for the prototype.

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2 Overall Description

This section outlines the prototype architecture description, prototype functional description, and external interfaces.

2.1 Product Perspective

BevOS is a unified operational platform for beverage manufacturers, centralizing workflow management, approval tracking, production scheduling, document management, and logistics coordination. It improves visibility, reduces manual effort, and creates a scalable foundation for growth by eliminating the data fragmentation caused by spreadsheets and email chains and by automating quotes, purchase orders, approvals, and production schedules.

The system runs on standard cloud or local server infrastructure, with application and database containers supported by redundant infrastructure so the platform tolerates partial functionality during minor network outages, and is reached from ordinary desktop and mobile browsers.

It is built in Python with Django as the web framework, PostgreSQL for the data layer, and Docker and Docker Compose for containerization, with Pytest and GitHub Actions supporting testing and CI/CD. As shown in the MFCD in Figure 2, users work through a user-facing interface while BevOS core services supply the business logic across quotes, communications, logistics, document storage, reports, and the database.

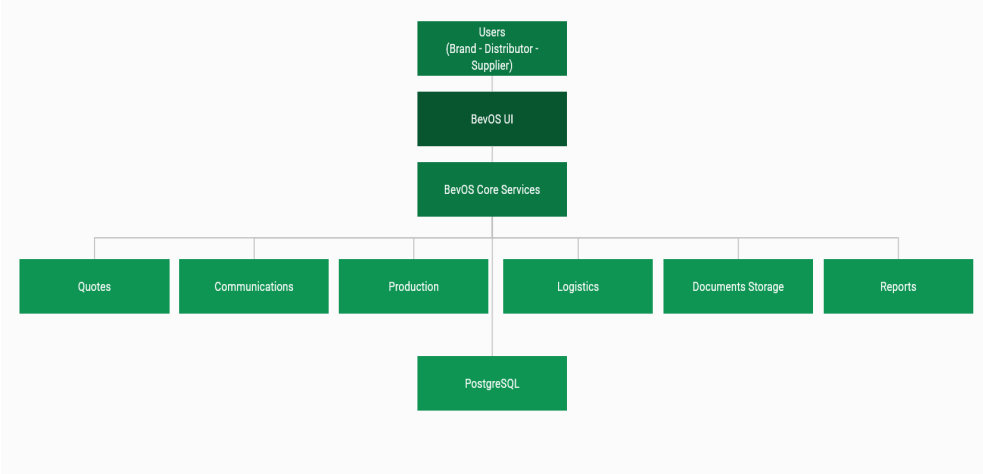


Figure 1 – MFCD Structure

2.2 Product Functions

Workflow management centralizes all operational activity from concept to store shelves, replacing disconnected spreadsheets and email chains with a single process pipeline. Product request management streamlines product and artwork requests through automated generation and approval routing, escalating to partial automation where full automation is inappropriate. The quote system calculates quotes automatically and supports supplier negotiation and distributor pricing tracking, while approval tracking routes manual and automated approvals for artwork, purchase orders, and production schedules so items do not stall.

Production management coordinates manufacturing and scheduling with co-packers and internal teams, and logistics coordination handles carrier coordination, shipping updates, and delivery tracking from facility to destination. Supporting these are a secure document repository for specifications and legal and technical files, a shared communications channel that prevents lost updates, dashboards and reports that turn operational data into real-time business insight, and an issue tracker that routes internal user feedback into a structured pipeline for resolving emerging shortcomings.

2.3 User Characteristics

Brands, clients, and distributors initiate product requests, track production milestones, approve artwork, and oversee project lifecycles. Suppliers and co-packers receive purchase orders, respond to quotes, and coordinate manufacturing requirements. Logistics providers and carriers manage shipping coordination, transport tracking, and delivery updates. Operations and finance teams monitor production schedules, manage invoicing, analyze performance metrics, and review reports. BevOS developers and IT maintainers handle system administration, bug tracking and ticket queues, and platform scalability. Access is granted by role, so each user sees only the information and actions their responsibilities require.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A